

NEET Important Chapter – Electromagnetic Waves

Important Concepts of Electromagnetic Waves

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In the previous chapters, we have learnt that electric current produces magnetic fields and changing magnetic fields produce electric fields.

In this chapter, we will discuss whether changing electric fields develop magnetic fields or not. James Clerk Maxwell explained that not only electric current but also varying electric fields, with time, also generate magnetic fields. He introduced displacement current along with conduction current.

We will also discuss Maxwell's equations in this chapter, which prove the existence of electromagnetic waves. Maxwell corrected the ampere law by adding the effect of displacement current and gave Ampere – Maxwell's law.

We will also get to know what is the source of electromagnetic waves and their nature, speed, frequency, wavelength and intensity in this chapter, as well as, know about the relationship between electric field and magnetic field.

This chapter also discusses the spectrum of electromagnetic waves, which is an electromagnetic spectrum that contains a list of waves like radio wave, microwave, infrared wave, visible wave, and ultra visible rays.

We will also deal with light, which is an electromagnetic wave and find the relationship between electric field and magnetic field.

Now, let's move onto the important concepts and formulae related to NEET exams along with a few solved examples and previous year questions.

Important Topics of Electromagnetic Waves

- Revision Notes of Electromagnetic waves
- Electromagnetic Spectrum
- Electromagnetic Waves – Definition, Equation and Properties
- NCERT Solutions – electromagnetic wave
- Displacement equation
- Maxwells Equations
- Radio Waves
- The Electromagnetic Spectrum Visible Light
- Velocity of Electromagnetic waves

Electromagnetic Waves Important Concept for NEET

Name of the Concept	Key Points of the Concepts
1. Conduction current	• The current carried by a conductor due to the flow of charges is called conduction current. • Outside the capacitor plate, we only have conduction current.

2. Displacement current	• The current which is due to the changing electric field is called displacement current. • It is due to the time varying electric field. • Inside the capacitor, there is no conduction current, there is only displacement current.
3. Source of Electromagnetic waves	• Accelerated charges radiate electromagnetic waves. The frequency of electromagnetic waves naturally equals the frequency of oscillation of charge.

4. Nature of Electromagnetic waves	• Electromagnetic waves propagate in the direction of a propagation vector that is k. It is perpendicular to both electric and magnetic fields. Electric fields and magnetic fields are also perpendicular to each other. $K = \frac{2\pi}{\lambda}$ Here λ is the wavelength of em wave. • Electromagnetic waves are transverse in nature.
5. Intensity	• Power per unit area or energy crossing per unit area per unit time perpendicular to the direction of propagation is called intensity of a wave.

6. Electromagnetic spectrum

- According to frequency, all electromagnetic waves have to be classified. This is known as the electromagnetic spectrum.
- In the Electromagnetic spectrum, all the different regions are known by different names like gamma-rays, X-rays, ultraviolet rays, visible rays, infrared rays, microwaves and radio waves.
- The range of wavelengths in the visible ray is 400 nm to 700 nm.

7. Electromagnetic pressure

- As we know, electromagnetic waves carry energy and

7. Electromagnetic pressure	momentum. An em wave exerts pressure, called radiation pressure.
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List of Important Formulae for Electromagnetic Waves

S.No.	Name of the Concept	Formula
1.	Ampere's circuital law	Here $i(t)$ is the time-dependent current which is flowing through the circuit. μ_0 is permeability in vacuum.
2.	Displacement current	— Here $\frac{d\phi_E}{dt}$ is electric flux and ϵ_0 is permittivity in vacuum.
3.	Speed of propagation of wave	— Here ω is angular frequency and k is propagation vector $K = \frac{2\pi}{\lambda}$ Here λ is the wavelength of em wave. Angular frequency = $2\pi f$ Here f is the frequency of wave
4.	Relation between frequency and wavelength	$c = \lambda f$ Here c is the velocity of light.
5.	Relation between electric and magnetic field	— Here c is the velocity of light that has a constant value 3×10^8 m/s

6.	First Maxwell's equation (Gauss law for electricity)	—
7.	Second Maxwell's equation (Gauss law for Magnetism)	
8.	Third Maxwell's equation (Faraday's law)	—

9.	Fourth Maxwell's equation (Ampere Maxwell law)	— Here ϵ_0 is permittivity in vacuum. And μ_0 is the permeability in vacuum.
10.	Velocity of light in terms of permittivity and permeability	$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$
11.	Electromagnetic pressure	$P = \frac{U}{c}$ Here U = total energy and c is light velocity

Solved Examples

1. A plane electromagnetic wave of frequency 4 Mhz travels in free space along the x-direction. At a particular point in space and time, $E = 6.3 \text{ V/m}$. What is B at this point?

Sol:

Given:

Electric field = $E = 6.3 \text{ V/m}$

We know that velocity of light = $c = 3 \times 10^8 \text{ m/s}$

The relation between electric and magnetic fields is $E = cB$

$$B = \text{————} =$$

For direction - As per question, the electric field is in y-direction and waves propagate in x-direction. So the magnetic field is perpendicular to both the wave vector (K) and the electric field (E).

Therefore, the magnetic field is along the z-direction

Key Point - Keep in mind the relation between electric and magnetic fields.

2. A charged particle oscillates about its mean equilibrium position with a frequency of Hz. What is the frequency of the electromagnetic waves produced by the oscillator?

Sol:

The frequency of the electromagnetic wave is the same as that of an oscillating charged particle about its equilibrium position, which is Hz.

Key Point: As accelerated charges radiate electromagnetic waves, the frequency of electromagnetic waves naturally equals the frequency of the oscillation of charge.

Previous Year Questions From NEET Paper

1. The ratio of contributions made by the electric field and magnetic field components to the intensity of an electromagnetic wave is (c is speed of electromagnetic wave) (NEET 2021)

a. C : 1

b. 1 : 1

c. 1 : c

d.

Sol:

The intensity of an electromagnetic wave is given by:

The average electric energy = —

The average magnetic energy = —

We know that the relation - —

—

Here, c is the velocity of light that has a constant value.

The average electric energy is —

= —

As we can see that the energy due to electric fields and magnetic fields is the same, therefore they make the same contribution in intensity.

Trick - Average energy due to electric field and magnetic field are equal.

2. For a plane electromagnetic wave propagating in the x-direction, which one of the following combinations give the correct possible direction for electric field (E) and magnetic field (B), respectively ?(NEET 2021)

1.

2.

3.

4.

Sol:

Option (4) is correct.

As per the question, the waves propagate in the x-direction. So the wave vector ($\vec{k}$) is also in x-direction. The magnetic field and electric field both are perpendicular to the wave vector ($\vec{k}$).

Direction of propagation of electromagnetic waves or k vector is along the product of electric or magnetic field.

By checking all the options, it can be seen that only option (c) follows the condition.

Trick – Electromagnetic waves propagate in the direction of the propagation vector and perpendicular to magnetic field and electric field.

Practice Questions

1. A plane electromagnetic wave travels in vacuum along the z -direction. What can you say about the directions of its electric and magnetic field vectors? If the frequency of the wave is 300 MHz, what is its wavelength? (**Ans: 1m**)
2. A radio can tune into any station in the 7.5MHz to 12MHz band. What is the corresponding wavelength band? (**Ans: 40m to 25m**)

Conclusion

In this article, we have provided important information regarding the chapter, such as important topics, concepts, and formulae, etc. Students should work on more solved examples for securing good marks in the [NEET 2022](#) exams. Try to remember the relationship between all the parameters that are required to understand the electromagnetic wave.

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